Abinaya Gowthaman

Python Developer | Generative AI

Manamadurai, Sivagangai | +91 9384755020 | abinayagowthaman004@gmail.com | linkedin.com/in/abinaya-g-bb00a5327

PROFILE

Computer Science undergraduate focused on Python development and Generative AI, with hands-on experience in FastAPI, RAG pipelines, LLM integration, and AI-powered applications. Strong in problem-solving and backend development, with a focus on building practical, scalable solutions.

PROJECTS

TN Scheme AI – Government Scheme Assistant

FastAPI, Python

- Engineered a Retrieval-Augmented Generation (RAG) pipeline using Python, FastAPI, and Google Gemini for government scheme query resolution.
- Implemented semantic retrieval with ChromaDB to fetch relevant scheme-document chunks and generate context-grounded responses with source information.
- Integrated JWT-based authentication and React frontend with FastAPI REST endpoints; deployed services on Render and Vercel.

https://tn-scheme-ai.vercel.app | https://github.com/Abinaya26-G/TN-Scheme-AI

RepoChat AI – GitHub Repository Assistant

Python, FastAPI, React, PostgreSQL, pgvector, Google Gemini

- Developed a full-stack AI-powered GitHub Repository Assistant using Python, FastAPI, and React, enabling developers to interact with repository source code through natural-language queries.
- Implemented a RAG pipeline with automated code parsing, chunking, Gemini embeddings, and PostgreSQL (pgvector) to perform semantic search and retrieve relevant code context for AI-generated responses.
- Integrated JWT-based authentication and REST APIs, and deployed the application using Render and Vercel for production accessibility.

https://repochat-ai-one.vercel.app | https://github.com/Abinaya26-G/repochat-ai

SKILLS

- **Programming Languages:** Python, Java, SQL
- **Backend Development:** FastAPI, REST APIs, Spring Boot
- **Generative AI:** RAG, LLM Integration, Prompt Engineering
- **Vector Databases:** ChromaDB, PostgreSQL, pg vector
- **Tools & Cloud:** Git, GitHub, Docker, AWS, Jenkins, Postman

INTERNSHIP

Full Stack Web Development Intern | iTECH Academy, Tirunelveli

May 2026 - June 2026

- Developed FastAPI backend endpoints for a document-based question-answering application.
- Implemented document chunking and embedding generation, storing vectorized text chunks in a vector database for semantic retrieval.
- Integrated retrieved document context into a RAG pipeline to generate context-aware answers to user queries.

EDUCATION

B.E - Computer Science and Engineering

2023 - 2027

AAA College of Engineering and Technology